

# MBTI Personality Type Classification from User-Level Text

Kontos Christos\*, Efthymiadis Konstantinos\*\*

\* School of Informatics, Aristotle University of Thessaloniki, Greece  
ckontosb@csd.auth.gr

\*\* School of Informatics, Aristotle University of Thessaloniki, Greece  
kefthyme@csd.auth.gr

## ABSTRACT

This paper investigates Myers-Briggs Type Indicator (MBTI) personality prediction from user-level online text using the Kaggle MBTI dataset as the supervised benchmark, with the task being represented through four binary axes, namely Extraversion versus Introversion, Intuition versus Sensing, Thinking versus Feeling, and Judging versus Perceiving, with a complete type being reconstructed by combining the four predicted labels. The study compares Term Frequency, Inverse Document Frequency (TF-IDF) models with Logistic Regression, calibrated Linear Support Vector Classifier (LinearSVC), Classifier Chains, and Label Powerset learning, while also evaluating shared encoder transformer models based on Bidirectional Encoder Representations from Transformers (BERT) architectures, such as MentalRoBERTa, RoBERTa, and ModernBERT. Moreover, trained MBTI probability outputs are tested on Pandora to examine whether the learned axis scores transfer to related Big Five OCEAN traits, namely Openness, Conscientiousness, Extraversion, Agreeableness, and Neuroticism.

The strongest result is achieved by TF-IDF Label Powerset, which obtains the highest macro-F1 along with the highest exact type accuracy among the evaluated models, with MentalRoBERTa and RoBERTa-base remaining competitive, however not outperforming the strongest sparse lexical baseline. In addition, the Pandora transfer analysis shows weak but interpretable relationships between predicted MBTI axis scores and related OCEAN traits, especially for Extraversion. Therefore, the results suggest that MBTI models capture some personality related language signals, but the findings should be interpreted as benchmark performance on self reported labels rather than as validated personality measurement.

### *Artifact Availability:*

The source code, data, and other artifacts have been made available at <https://github.com/cdkontos/MBTI-Personality-Type-Classification>.

## I. INTRODUCTION

The rapid growth of user-generated text on online platforms has created new opportunities for studying how people express personal characteristics through language, since forum posts, social media comments and online discussions often contain repeated patterns in word choice, emotional expression, topics of interest, pronoun use and interaction style. Previous work in computational personality recognition has shown that linguistic cues can be useful for estimating personality-related variables from text (Mairesse et al., 2007), although personality prediction remains a sensitive and uncertain task, which means that it should be approached carefully and evaluated as a modelling problem rather than as a direct measurement of a person. As such, the purpose of this work is not to create a decision-making system, but to examine how different natural language processing models behave on a controlled personality classification benchmark.

This study focuses on the Myers-Briggs Type Indicator (MBTI), which is a widely known personality type framework that describes personality through four binary categories, namely Extraversion versus Introversion, Intuition versus Sensing, Thinking versus Feeling and Judging versus Perceiving (Myers et al., 1998). These four axes combine in order to form sixteen possible personality types, such as INFP, INTJ and ENTP, and this structure makes MBTI prediction an interesting classification problem from a machine learning perspective, as it can be approached in multiple ways. A model could predict one of the sixteen complete personality types directly, it could predict each of the four axes separately, or it could treat the task as a multilabel problem where all four axes are learned together and then combined into a final MBTI type (Zhang & Zhou, 2014).

Predicting MBTI from text is challenging because the label belongs to the author rather than to any individual sentence or post, which means that a single piece of text may be short, sarcastic, highly topic specific or unrepresentative of the user's broader

writing style and, as a result, this project treats each user as the unit of analysis and classifies aggregated user level text rather than isolated posts, something which much better matches the way the labels are assigned in the dataset. The task is also complicated by the fact that the Kaggle MBTI dataset used in this work contains self reported personality labels (Jolly, 2017), meaning that the labels may be noisy and should not be interpreted as professional psychological assessments. In addition to that, online MBTI datasets may contain explicit label leakage, because users can directly mention their own type in the text, therefore the preprocessing pipeline removes any explicit MBTI type mentions before training, so that models are encouraged to learn broader linguistic patterns instead of simply memorizing leaked labels. Another important difficulty of this classification problem is the class imbalance that exists between types, as some MBTI types and axes appear much more frequently than others, which can make ordinary accuracy misleading when a model performs well on common classes but fails on less frequent ones. For this reason, this paper mainly evaluates models using axis level macro-F1, while also reporting Receiver Operating Characteristic, Area Under the Curve (ROC-AUC), balanced accuracy and exact full type accuracy, so that performance can be examined from several perspectives. This evaluation setup makes it possible to compare the quality of the four individual axis predictions, the ability of each model to rank examples by confidence and the stricter requirement of reconstructing all four MBTI letters correctly.

The project compares two broad types of approaches, with the first being classical models based on TF-IDF word and character n-grams (Salton & Buckley, 1988), which are combined with multilabel learning strategies such as Binary Relevance, Classifier Chains and Label Powerset classification (Read et al., 2011; Tsoumakas et al., 2011; Zhang & Zhou, 2014). These models provide strong baselines because lexical and stylistic features are often highly informative in text classification tasks, especially when the dataset contains repeated expressions, community specific vocabulary and stable writing patterns. The second type of approach consists of shared encoder transformer models, where a pretrained language model is fine tuned with four output heads, where each one corresponds to one of the four MBTI axes, something which allows the same contextual representation to support all four predictions (Devlin et al., 2019; Ji et al., 2022; Liu et al., 2019; Warner et al., 2025). As such, by comparing these two modelling families, this work investigates whether contextual neural representations provide a meaningful advantage over

sparse lexical models for user level personality prediction.

In addition to supervised evaluation on the Kaggle MBTI dataset, the report includes an external zero-shot analysis using the Pandora Reddit personality dataset (Gjurkovic et al., 2021), which is used to examine whether the learned MBTI signals show any relationship with a different personality framework. Pandora is not used as an MBTI labelled test set, but instead trained MBTI models are applied to the Pandora user text, and their predicted MBTI axis probabilities are compared with theoretically related OCEAN traits (Goldberg, 1990; McCrae & Costa, 1989). This analysis provides an exploratory check of whether the learned MBTI signals transfer beyond the original dataset and whether they align with broader personality dimensions such as extraversion, openness, conscientiousness and agreeableness.

The rest of this paper follows the same structure as the experimental workflow, since Section II reviews related work on computational personality recognition, MBTI and Big Five personality modelling, multilabel classification and transformer-based text encoders, while Section III describes the datasets, preprocessing pipeline and evaluation setup. Section IV then presents the classical and transformer-based modelling approaches, Section V reports and discusses the experimental results, including supervised MBTI performance and Pandora transfer analysis, and Section VI concludes the paper by summarizing the main findings and outlining possible directions for future work.

## II. RELATED WORK

### A. Personality Prediction from Language

Personality type prediction is based on the observation that language use is connected to stable individual differences, since people often show recurring patterns in word choice, emotional expression, topic preference and interaction style. Earlier work showed that linguistic cues can be used to predict Big Five personality traits from both conversation and written text (Mairesse et al., 2007), and later research expanded this idea to large scale social media settings, where language samples are easier to collect and can be connected to self reported personality information.

One major direction in this area uses closed vocabulary analysis, where psychological categories are measured through predefined dictionaries such as Linguistic Inquiry and Word Count (LIWC) (Pennebaker et al., 2015). Closed vocabulary features are useful because they are interpretable, as researchers know which word categories are being counted, but they can also miss slang, topic specific

phrasing and new expressions that appear in online communities. As such, open vocabulary methods address this limitation by allowing words, phrases and topics to emerge from the data itself, something which makes them especially useful for studying informal online language.

Large scale social media research has shown that open vocabulary language analysis can be connected to personality, gender and age (Schwartz et al., 2013), with further work also showing that social media language can be used to build personality assessments that correlate with self reports and informant reports (Park et al., 2015).

### *B. MBTI and Big Five Personality Frameworks*

The Big Five model treats personality as five continuous traits, that are, namely, openness, conscientiousness, extraversion, agreeableness and neuroticism (Goldberg, 1990) and it is widely used in psychological research, with extensive empirical support, while MBTI is different because it divides personality into four binary preferences and combines them into sixteen types (Myers et al., 1998). This categorical structure is attractive for classification experiments because it creates clear output labels, although it also simplifies continuous personality variation and can make boundary cases difficult to interpret.

Prior psychological research has examined how MBTI dimensions relate to Big Five traits, showing that some MBTI axes map more clearly to Five Factor dimensions than others (McCrae & Costa, 1989). Extraversion is the most direct relationship, while Intuition is often associated with openness and Judging is often associated with conscientiousness. Thinking and Feeling have a looser relationship with agreeableness, which means that this mapping should be treated as approximate rather than exact. These expected relationships motivate the Pandora transfer evaluation in this report, where MBTI model outputs are compared with related OCEAN traits.

### *C. Social Media Personality Datasets*

The Kaggle MBTI dataset is a popular benchmark for MBTI prediction because each row contains a user type together with a collection of recent posts, which makes it useful for comparing models on user level personality classification, although it should not be treated as a professionally controlled psychological dataset (Jolly, 2017). Since the labels are self-reported, the text comes from online communities and the distribution of personality types is highly uneven, the conclusions drawn from this dataset should be framed as dataset specific predictive performance rather than as evidence of definitive psychological diagnosis.

Other datasets have also been proposed for personality and author profiling, with one multilingual Twitter corpus containing users annotated with MBTI type and gender, which shows that MBTI style author profiling can be studied beyond English only settings (Verhoeven et al., 2016). In addition to that, the Pandora dataset provides Reddit comments with personality and demographic labels, including Big Five annotations for a subset of users, making it especially relevant for this work because it allows a cross framework comparison between MBTI predictions and OCEAN traits, rather than only another direct evaluation on the same MBTI dataset (Gjurkovic et al., 2021). Since personality datasets are built from personal language and self-reported traits, their use requires caution, especially in social media settings where text may contain information beyond the labels that were originally collected. Prior research has shown that digital records can be used to infer sensitive personal attributes, which makes it important to frame personality prediction as a limited research task rather than as a reliable tool for judging individuals (Kosinski et al., 2013).

### *D. Classical Text Classification and Multilabel Learning*

Traditional text classification methods remain strong baselines for personality prediction because online language often contains repeated lexical patterns, topic markers and stylistic cues, which can be captured effectively through sparse feature representations. TF-IDF is commonly used for this type of representation because it weights terms according to their frequency in a document and their rarity across the corpus, allowing common but less informative words to contribute less than more discriminative terms (Salton & Buckley, 1988). This makes it suitable for user level text classification, especially when the goal is to represent both repeated writing patterns and words that distinguish one personality label from another.

MBTI prediction can also be treated as a multilabel problem because each user can be represented through four binary axis labels, one for each MBTI dichotomy. A simple approach is to train one independent classifier for each axis, although this ignores possible dependencies between the four labels, while a more structured approach is to model relationships between labels through chained predictions (Read et al., 2011). Another approach is to transform the multilabel task into a multiclass problem over unique label combinations, which is especially natural for MBTI because the four binary axes create exactly sixteen possible full personality types (Tsoumakas et al., 2011).

Recent work on MBTI prediction has started to treat the task less as a simple sixteen type classification problem and more as a set of related decisions across the four personality axes. In one example, explainable multilabel models were tested on Reddit posts and the Kaggle MBTI dataset, so that the predictions could be connected back to the textual features that influenced them (Kong & Sokolova, 2024). Domain adapted encoders are also useful in this area because personality related text often contains personal experience, emotion, opinion and informal expression, rather than the more polished language found in many standard pretraining corpora. MentalBERT and MentalRoBERTa were developed for mental healthcare related Natural Language Processing (NLP) tasks, which makes them relevant as comparison models when working with personal online text (Ji et al., 2022). Although MBTI prediction is not a mental health detection task, MentalRoBERTa is still worth evaluating because its pretraining domain is closer to the kind of self-expression, emotional content and informal discussion that can appear in online personality datasets.

#### *E. Transformer Encoders and Domain Adaptation*

Transformer based encoders are relevant to personality prediction as they can represent words in context, which allows the model to use surrounding language rather than relying only on independent word counts. This is useful for user level social media text, since online posts often contain informal phrasing, personal disclosure, slang and fragmented expression, which can be difficult to capture with surface level features alone. However, because this type of language differs from standard written corpora, domain adaptation becomes important, and previous work has shown that continued pretraining on domain data or task data can improve downstream performance (Gururangan et al., 2020).

This is especially relevant for personality related text, as users often express emotions, opinions and personal experiences in ways that overlap with psychological and social media language. MentalBERT and MentalRoBERTa are examples of pretrained language models designed for mental healthcare related NLP tasks, which makes them useful points of comparison when working with personal online text (Ji et al., 2022). Although MBTI prediction is not a mental health detection task, MentalRoBERTa is still relevant because it has been adapted to language that often contains self-expression, emotional content and informal discussion, all of which may also appear in online personality datasets.

Long context encoders provide another direction for user level personality prediction, because each user may have many posts and a single short text may not be representative of their broader writing style. ModernBERT was designed as a bidirectional encoder for efficient long context fine tuning and inference, making it a relevant model for testing whether access to more user text can improve personality classification (Warner et al., 2025). In principle, a longer context window should help the model inspect a wider sample of a user’s writing, although the benefit still depends on optimization, data quality and how the text is selected during training and evaluation.

### III. DATASETS AND PREPROCESSING

#### *A. Kaggle MBTI User-Level Dataset*

The supervised experiments use the Kaggle Myers-Briggs Personality Type dataset, where each instance contains a four-letter MBTI type with a block of 50 online posts that are associated with one user (Jolly, 2017). The data is processed at user level because the MBTI label describes the author rather than any single post, and this makes the split design important, since separating individual posts could allow text from the same author to appear in both training and testing data. As such, the experiments use user level partitions, so that each author appears in only one split and the evaluation better reflects prediction on new users.

The full MBTI type is converted into four binary labels that correspond to the four MBTI axes defined by the personality framework (Myers et al., 1998). The EI axis represents Extraversion versus Introversion, the NS axis represents Intuition versus Sensing, the TF axis represents Thinking versus Feeling, and the JP axis represents Judging versus Perceiving. As such, for each axis, the positive class is defined as the first letter in the axis name, namely E, N, T and J, which allows the model to predict four probabilities and then reconstruct a full MBTI type by combining the predicted letters. This representation is summarized in Table 1.

The cached split contains 6,071 training users, 1,301 validation users and 1,301 test users, which follows an approximate 70, 15 and 15 split, and the partitions are stratified by the complete MBTI type rather than by only one axis. Stratification is important because the dataset is imbalanced, and without it, rare personality types could be weakly represented or even missing from the validation and test partitions.

The text cleaning pipeline removes URLs, converts repeated post separators into spaces, normalizes whitespace and removes explicit MBTI type mentions, so that the model cannot simply learn from

leaked type labels in the text. For the classical models, the cleaned text is lowercased and restricted to alphabetic characters, which fits the TF-IDF pipeline, while for the transformer models, a separate text version is used that keeps more of the original casing and converts post separators in a way that better preserves the flow of the user history, while still removing explicit type mentions. Since many users have more text than a transformer encoder can process at once, user histories are divided into chunks for transformer evaluation, and the resulting chunk predictions are averaged back to the user level. This separation is useful because sparse lexical models and tokenized transformer encoders use text in different ways, while the final evaluation remains at the author level.

| <i>Axis</i> | <i>Positive</i> | <i>Negative</i> | <i>Meaning</i>                |
|-------------|-----------------|-----------------|-------------------------------|
| <i>EI</i>   | E               | I               | Extraversion vs. Introversion |
| <i>NS</i>   | N               | S               | Intuition vs. Sensing         |
| <i>TF</i>   | T               | F               | Thinking vs. Feeling          |
| <i>JP</i>   | J               | P               | Judging vs. Perceiving        |

Table 1: Binary MBTI axis representation used by the multilabel models.

#### B. Pandora OCEAN Transfer Dataset

The Pandora dataset is used for external transfer analysis rather than for supervised MBTI training (Gjurkovic et al., 2021), as it contains Reddit users with personality and demographic annotations, including Big Five and OCEAN traits for part of the dataset. Because Pandora does not provide direct MBTI labels for this evaluation, it is not used as a second MBTI accuracy test, but, instead, the trained MBTI models are applied to Pandora user text in order to produce MBTI axis probabilities, and these probabilities are then compared with related OCEAN traits.

The transfer mapping is based on common theoretical relationships between MBTI and the Big Five framework (Goldberg, 1990; McCrae & Costa, 1989), with the probability of Extraversion, P of E, being compared with Big Five extraversion, while the probability of Intuition, P of N, being compared with openness. The probability of Judging, P of J, is compared with conscientiousness, and the probability of Feeling, P of F, is compared with agreeableness. Since the TF model outputs the probability of Thinking, P of F is calculated as one minus P of T. Neuroticism is not assigned a direct MBTI counterpart, as the four MBTI axes do not contain a clear emotional stability dimension.

The Pandora dataset text is processed with the same general user level approach used for the supervised MBTI data, with explicit MBTI type strings being removed, empty text avoided, and very short

examples being filtered out before evaluation. For transformer models, long user histories are split into chunks and predictions are averaged back to the user level, so that the transfer analysis remains consistent with the supervised author level design.

#### C. Data Quality Considerations

The two datasets differ in both label construction and platform context, which makes them useful for different parts of the evaluation, as the Kaggle MBTI dataset is based on self declared type labels and mixed online posts, while Pandora is based on Reddit language and Big Five trait annotations. These differences make Pandora useful for checking whether MBTI model scores show any relationship with another personality framework, although they also mean that the two datasets cannot be compared as if they measured the same label space.

The cleaning process reduces direct label leakage by removing explicit MBTI type names, but it cannot remove every indirect cue, as users may still discuss MBTI related topics without naming a specific type, or they may use community specific vocabulary that is strongly associated with one label. Therefore, the results should be interpreted as benchmark performance on user level text data, rather than as evidence that the models fully understand personality in a psychological sense.

### IV. MODELS AND FINE-TUNING

#### A. TF-IDF Feature Representation

The classical models represent each user through a feature union of word based and character based TF-IDF vectorizers (Salton & Buckley, 1988), so that the same user text can be described both through its vocabulary and through smaller stylistic patterns. The word vectorizer uses unigrams and bigrams, with a minimum document frequency of two, a maximum document frequency of 0.95 and a maximum of 120,000 features, while the character vectorizer uses word boundary character n-grams from length three to five, with a minimum document frequency of two and a maximum of 80,000 features. In both feature spaces, sublinear term frequency scaling is applied, which allows repeated terms to remain informative without dominating the representation through volume.

This combined representation is useful because MBTI prediction from online posts may depend not only on what users discuss, but also on how they write. Word n-grams can capture repeated topics, expressions and vocabulary choices, while character n-grams can capture smaller cues such as spelling habits, abbreviations, word fragments and informal stylistic patterns. As a result, the TF-IDF pipeline

provides a strong classical baseline, as it can represent recurring lexical and stylistic differences efficiently before any more complex neural model is introduced.

### *B. Binary Relevance Baselines*

The first baseline uses Binary Relevance, where one Logistic Regression classifier is trained for each MBTI axis (Cox, 1958; Zhang & Zhou, 2014). Although all four classifiers use the same TF-IDF feature matrix, each axis is handled as a separate binary problem, something which makes it possible to tune the regularization strength, class weighting and decision threshold depending on the distribution of that axis. This is important as the four MBTI axes are imbalanced and not of the same difficulty, and, as such, a threshold that works well for one axis may not be suitable for another.

The second Binary Relevance baseline uses calibrated LinearSVC models (Cortes et al., 1995), which provide an additional strong sparse text classification approach. Linear Support Vector Machines are effective in high dimensional feature spaces, but their raw outputs are decision scores rather than probabilities, so sigmoid calibration is applied before evaluation. This allows ROC-AUC to be calculated and threshold tuning to follow the same logic as the Logistic Regression baseline, meaning that both can be compared through ranking quality and final axis level decisions.

### *C. Classifier Chains and Label Powerset*

The Classifier Chain ensemble extends Binary Relevance by allowing relationships between MBTI axes to influence prediction (Read et al., 2011). Instead of predicting all four labels independently, each chain predicts the axes in a sequence, where earlier predictions can provide additional information for later ones. As such, in order to avoid depending too strongly on a single arbitrary axis order, seven chains are trained with randomized label orders and their predicted probabilities are then averaged, something that keeps the model close to the Binary Relevance setup while still allowing for dependencies between labels to affect the final output.

The Label Powerset model uses a different strategy because it transforms the four binary axis labels into one complete MBTI type (Tsoumakas et al., 2011). The four labels are encoded as a single integer class from 0 to 15, and a multiclass classifier is trained to predict this full type directly. This is a natural fit for MBTI because the label space is small and structured, since the four binary axes produce sixteen possible type combinations, and therefore the model can learn

patterns that belong to complete types rather than learning each axis separately.

### *D. Shared-Encoder Transformer Architecture*

The neural models use a shared encoder setup, where one pretrained transformer processes the input text, and four binary classification heads produce the MBTI axis predictions. This follows the general fine-tuning approach used with pretrained encoder models, where a contextual language representation is passed to a task specific output layer (Devlin et al., 2019). In this case, the output layer is split into four heads, so that the model can learn one shared representation of user language while still producing separate decisions for Extraversion versus Introversion, Intuition versus Sensing, Thinking versus Feeling and Judging versus Perceiving.

Training uses binary cross entropy with logits because each user has four binary labels rather than one single class label. Furthermore, in order to address imbalance, positive class weights are computed from the training distribution, then square root scaling and clipping are applied so that minority classes receive more weight without making optimization unstable. After training, decision thresholds are selected on the validation set for each axis and then kept fixed for the final test evaluation, which ensures that threshold tuning does not use information from the test set.

Since many user histories are longer than a transformer input window, the text is split into chunks before being passed through the encoder. MentalRoBERTa and RoBERTa base use 260-word chunks, with up to three chunks per user during training and up to five chunks during evaluation, while the ModernBERT long ablation uses 500-word chunks, with up to two chunks per user during both training and evaluation. During evaluation, the probabilities from all available chunks are averaged back to the user level, so that the final prediction represents the user rather than one isolated chunk of text.

The transformer configurations that are used in this work include MentalRoBERTa, RoBERTa base and ModernBERT long, chosen in order to compare different encoder settings under the same user level prediction task. MentalRoBERTa represents a domain adapted encoder for personal and psychologically expressive language (Ji et al., 2022), RoBERTa base serves as a general-purpose encoder baseline (Liu et al., 2019), and ModernBERT long represents a newer long context configuration that can process larger portions of user text (Warner et al., 2025).

### E. Evaluation Procedure

The main evaluation metric is macro-F1 averaged across the four MBTI axes, since the dataset is imbalanced and the evaluation should not be dominated by the majority side of each dichotomy. In addition, ROC-AUC is reported as it measures how well a model ranks positive and negative examples before a final threshold is applied, while balanced accuracy provides another axis level view that accounts for uneven class distributions. Together, these metrics make it possible to examine both the quality of the model scores and final binary axis predictions.

Exact type accuracy is also calculated by combining the four predicted axis letters into a complete MBTI type and comparing it with the true type, with this metric being stricter than axis level evaluation as a prediction is counted as correct only when all four MBTI letters are correct at the same time. Moreover, for the Label Powerset model, direct labelset argmax accuracy is also reported because the model is trained as a sixteen-class classifier, which allows the direct full type prediction to be compared with the type reconstructed from thresholded axis probabilities.

Finally, bootstrap confidence intervals are used for macro-F1 and ROC-AUC, so that close differences between models can be interpreted more carefully than they would be from point estimates alone (Efron, 1979). The bootstrap procedure repeatedly resamples the held-out test predictions and recalculates the evaluation scores, producing an interval around each metric and giving a clearer view of how stable the reported results are.

## V. RESULTS AND DISCUSSION

### A. Overall Model Comparison

The final held out test results show that the strongest model is TF-IDF Label Powerset, which reaches a macro-F1 of 0.7811 and a macro ROC-AUC of 0.8738, outperforming the Binary Relevance baselines, the Classifier Chain ensemble and all transformer configurations. As shown in Table 2, this model also achieves the highest balanced accuracy and the highest threshold based exact type accuracy, which indicates that the full sixteen type structure of MBTI is useful for this dataset.

The remaining TF-IDF models perform closely to one another, with Logistic Regression reaching a macro-F1 of 0.7236, LinearSVC reaching 0.7262 and the Classifier Chain ensemble reaching 0.7269. Although the chain ensemble slightly improves over the independent baselines, it does not approach the Label Powerset result, suggesting that direct full type modelling is more effective here than modelling axis dependencies through chains. Moreover,

MentalRoBERTa and RoBERTa base remain competitive, with macro-F1 scores of 0.7520 and 0.7513, yet both remain below the Label Powerset classifier. In contrast, the ModernBERT long ablation performs substantially worse, with a macro-F1 of 0.5887 and a ROC-AUC of 0.6448, which likely reflects the limitations of the specific fine tuning setup rather than a general limitation of long context encoders.

| <i>Model</i>                      | <b>F1</b> | <b>ROC-AUC</b> | <b>Bal. Acc.</b> | <b>Exact</b> |
|-----------------------------------|-----------|----------------|------------------|--------------|
| <i>TF-IDF Label Powerset</i>      | 0.7811    | 0.8738         | 0.7814           | 0.5461       |
| <i>TF-IDF Logistic Regression</i> | 0.7236    | 0.8297         | -                | 0.4235       |
| <i>TF-IDF LinearSVC</i>           | 0.7262    | 0.8295         | -                | 0.4366       |
| <i>TF-IDF Classifier Chain</i>    | 0.7269    | 0.8280         | 0.7300           | 0.4197       |
| <i>MentalRoBERTa 4-head</i>       | 0.7520    | 0.8433         | 0.7518           | 0.4685       |
| <i>RoBERTa-base 4-head</i>        | 0.7513    | 0.8406         | 0.7502           | 0.4593       |
| <i>ModernBERT-long 4-head</i>     | 0.5887    | 0.6448         | 0.5903           | 0.2327       |

Table 2: Final held-out MBTI test results from the notebook outputs

The same pattern is also visible in Figure 1, where Label Powerset separates clearly from the other models in both macro-F1 and ROC-AUC, while the two RoBERTa based transformer models remain in the middle of the comparison.

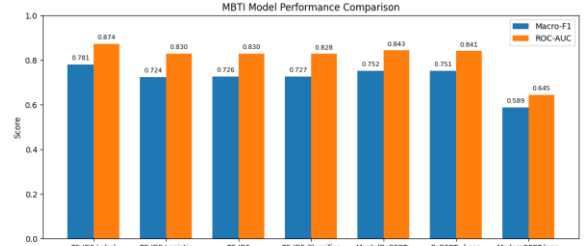


Figure 1: Macro-F1 and ROC-AUC comparison across all evaluated MBTI models

### B. Axis-Level Analysis

The axis level scores in Table 3 show that the TF axis is generally the easiest to classify, since the Label Powerset model reaches a macro-F1 of 0.8321 on this dimension, while MentalRoBERTa and RoBERTa base also exceed 0.81. The NS and JP axes are more difficult across most models, while EI falls between them depending on the model. This pattern suggests that Thinking versus Feeling may be associated with stronger lexical cues in the Kaggle data, whereas other MBTI preferences may be expressed more indirectly through writing style or topic choice.

EI is theoretically related to social orientation, but online text does not always reveal that distinction directly, since introverted users may write frequently online and extraverted users may produce similar text depending on the topic. Similarly, the JP axis may be difficult because planning, structure and spontaneity



are not always clearly visible in short or informal posts. Therefore, these axes may depend more on indirect and distributed cues, which makes them harder to classify consistently.

| <i>Model</i>                      | <b>EI</b> | <b>NS</b> | <b>TF</b> | <b>JP</b> |
|-----------------------------------|-----------|-----------|-----------|-----------|
| <i>TF-IDF Label Powerset</i>      | 0.7735    | 0.7689    | 0.8321    | 0.7497    |
| <i>TF-IDF Logistic Regression</i> | 0.7089    | 0.6796    | 0.8152    | 0.6906    |
| <i>TF-IDF LinearSVC</i>           | 0.7164    | 0.6769    | 0.8173    | 0.6942    |
| <i>TF-IDF Classifier Chain</i>    | 0.7129    | 0.6874    | 0.8094    | 0.6979    |
| <i>MentalRoBERTa 4-head</i>       | 0.7511    | 0.7190    | 0.8118    | 0.7261    |
| <i>RoBERTa-base 4-head</i>        | 0.7431    | 0.7192    | 0.8142    | 0.7287    |
| <i>ModernBERT-long 4-head</i>     | 0.5803    | 0.5126    | 0.7237    | 0.5383    |

Table 3: Axis-level macro-F1 scores for the main MBTI dimensions

The Label Powerset model improves all four axes compared with the independent TF-IDF baselines, which indicates that full type information helps not only exact type reconstruction, but also the individual axis decisions. This improvement is especially meaningful because a difficult axis value may become easier to predict when it appears as part of a more recognizable full type pattern.

### C. Exact MBTI Type Accuracy

Exact type prediction is much harder than axis level prediction because a full type is counted as correct only when all four letters are predicted correctly, and, as a result, errors can compound across axes, meaning that even models with reasonable axis level macro-F1 can produce lower exact type accuracy. This explains why every model has a much lower exact type score compared to its average per axis score.

The best exact type result is again obtained by TF-IDF Label Powerset, which reaches 0.5461 using thresholded axis predictions and 0.5515 using direct labelset argmax, slightly higher than the thresholded reconstruction result, which is expected because the model is trained to choose among complete type combinations. Meanwhile, MentalRoBERTa reaches 0.4685 and RoBERTa base reaches 0.4593, while the classical Binary Relevance baselines remain around 0.42 to 0.44. Figure 2 summarizes the thresholded exact accuracy results and shows the gap between the best full type model and the remaining approaches.

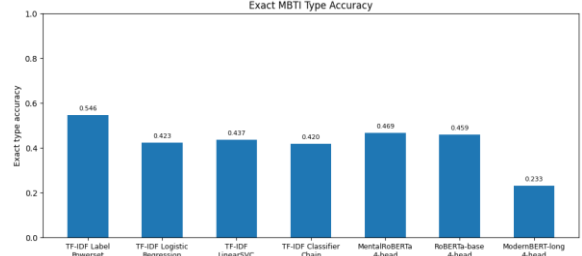


Figure 2: Exact MBTI type accuracy using thresholded axis predictions

### D. Why the Sparse Baseline Wins

The strong performance of TF-IDF Label Powerset may seem surprising at first, but it is reasonable for a moderate sized text classification task where labels are connected to repeated vocabulary, topics and community specific expressions. The Kaggle MBTI dataset likely contains many lexical signals that sparse features can capture efficiently, especially when full type combinations are modelled directly rather than separated into four independent axes.

Transformer models offer contextual representations, yet they also require careful optimization and enough training data to benefit from their larger capacity. In this experiment, user histories are split into chunks, and chunk probabilities are averaged, which may weaken information about how different posts relate across the full user history. Therefore, a future hierarchical model or attention based chunk aggregator could potentially use the user level structure more effectively than simple averaging.

Furthermore, these results also show why strong classical baselines are necessary, as, if only transformer-based models were compared with each other, MentalRoBERTa and RoBERTa base would appear to perform strongly, but the inclusion of Label Powerset changes the main conclusion. As such, for this benchmark, a well configured sparse lexical model is the strongest approach among the tested systems.

### E. Pandora OCEAN Transfer

The Pandora transfer analysis provides an external check of whether the trained MBTI models learn signals that have any relationship with a different personality framework. Since Pandora contains Big Five and OCEAN annotations rather than MBTI labels (Gjurkovic et al., 2021), it is not treated as another supervised MBTI test set, but, instead, the trained MBTI models are applied to Pandora user text, and their predicted axis scores are compared with theoretically related OCEAN traits. In particular, the probability of Extraversion is compared with Big Five extraversion, the probability of Intuition is compared with openness, the



probability of Judging is compared with conscientiousness, and the probability of Feeling is compared with agreeableness, since these links are commonly discussed when MBTI is interpreted through the Big Five framework (Goldberg, 1990; McCrae & Costa, 1989).

Figure 3 shows the Spearman correlations for all evaluated models across these expected MBTI to OCEAN links, with, overall, the correlations being positive but weak, and therefore the results suggesting only limited overlap between the learned MBTI scores and the related OCEAN traits. The strongest expected relationship appears between predicted Extraversion and Big Five extraversion, where RoBERTa-base reaches a Spearman correlation of 0.1114 and MentalRoBERTa reaches 0.1025. Moreover, this result is consistent with the theoretical connection between the two extraversion dimensions, even though the size of the correlation shows that the relationship is still modest, and, in addition, the Intuition to openness link produces a noticeable result, since ModernBERT-long reaches the highest correlation on this pair with a Spearman value of 0.0902, something which is interesting because, while ModernBERT-long performs poorly on the supervised MBTI test set, it still captures some language signal that aligns with openness in Pandora. Furthermore, the remaining expected links are weaker, since the best Judging to conscientiousness result is 0.0394 from TF-IDF Label Powerset, while the best Feeling to agreeableness result is 0.0550 from MentalRoBERTa. As such, the transfer analysis suggests that some personality related language patterns may carry across frameworks, but the connection is not strong enough to support treating MBTI predictions as substitutes for direct OCEAN measurement.

Therefore, the Pandora results should be interpreted as exploratory rather than confirmatory, as they show that the learned MBTI axis scores sometimes move in the expected direction when compared with related Big Five traits, but they also show that cross-framework transfer is limited. Consequently, when OCEAN labels are the real target, models should be trained and evaluated directly on OCEAN data instead of relying on MBTI predictions as an indirect replacement.

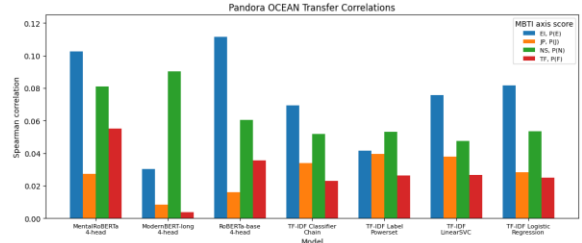


Figure 3: Spearman correlations between predicted MBTI axis scores and corresponding OCEAN traits in Pandora

#### F. Limitations, Ethics and Reproducibility

The main limitation is label quality, since the Kaggle MBTI labels are self-reported and may contain noise (Jolly, 2017). Furthermore, some users may mistype their personality, rely on an old test result, or identify with an online type community without formal testing, so the models trained on it should be understood as predicting dataset labels rather than validated psychological measurements. Moreover, although explicit MBTI strings are removed during preprocessing, residual leakage may still remain through indirect references to cognitive functions, type communities or personality related vocabulary and, as a result, the reported performance may partly reflect dataset specific cues rather than broader personality understanding.

Another limitation is platform bias, because both the Kaggle and Pandora data come from online communities and, as such, their language patterns may not generalize to workplace writing, academic writing or private communication. Therefore, models trained on these datasets may reflect demographic, cultural and platform specific biases, which is especially important because personality prediction is a sensitive form of inference, with work on digital records having shown that online behaviour can reveal private attributes (Kosinski et al., 2013). Therefore, systems that infer personality from text should not be used for hiring, admissions, insurance or other high stakes decisions, but, instead, research use should emphasize transparency, consent, data minimization and careful reporting of uncertainty. Finally, the experiments are made easier to reproduce through cached data splits, fixed random seeds, saved model outputs and the notebook code used to generate the reported tables and figures.

## VI. CONCLUSION

This work examined MBTI personality type classification from user level online text by comparing classical TF-IDF models, multilabel learning strategies and shared encoder transformer models. Overall, the results show that TF-IDF Label Powerset is the strongest approach, since it

outperforms the Binary Relevance baselines, the Classifier Chain ensemble and all transformer configurations. Therefore, the sixteen type structure of MBTI appears to be useful for this dataset, because modelling the complete type directly gives better performance than predicting each axis independently or only linking axes through classifier chains.

Consequently, the transformer models still provide useful comparisons, since MentalRoBERTa and RoBERTa base perform competitively and exceed the independent TF-IDF baselines on macro-F1, however, they do not outperform the Label Powerset model, which suggests that contextual neural representations are not automatically better for this task when the dataset is moderate in size and contains strong lexical patterns. In addition, the weaker ModernBERT long result shows that longer context alone is not enough to improve user level personality prediction, especially when chunking, fine tuning and aggregation choices may strongly affect performance, with the axis level results also showing that the task is uneven across MBTI dimensions. In particular, the Thinking versus Feeling axis is the easiest to classify, while the Intuition versus Sensing and Judging versus Perceiving axes are generally more difficult.

Moreover, exact type accuracy is much lower than axis level performance because all four letters must be predicted correctly at the same time, thus resulting in exact MBTI type prediction remaining a demanding evaluation setting, even when individual axis predictions are reasonably strong.

The Pandora transfer analysis provides an additional external check by comparing predicted MBTI axis scores with related OCEAN traits. The correlations are positive in the expected directions, especially for Extraversion and Big Five extraversion, however the values are weak.

Consequently, the transfer results should be interpreted as exploratory evidence of limited cross-framework overlap rather than as validation that MBTI predictions can replace direct Big Five measurement.

In conclusion, the main finding is that a well designed sparse lexical model can outperform more complex transformer models on this MBTI benchmark, and, therefore, future work should continue to include strong classical baselines, while also exploring better user level neural architectures, such as hierarchical encoders or attention based chunk aggregation. Furthermore, future experiments should evaluate stricter leakage removal, stronger cross-dataset testing and more careful treatment of self reported labels, since personality prediction from text is sensitive and should be reported with clear limitations.

## REFERENCES

- Cortes, C., Vapnik, V., & Saitta, L. (1995). Support-vector networks. *Machine Learning* 1995 20:3, 20(3), 273–297. <https://doi.org/10.1007/BF00994018>
- Cox, D. R. (1958). The Regression Analysis of Binary Sequences. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 20(2), 215–232. <https://doi.org/10.1111/J.2517-6161.1958.TB00292.X>
- Devlin, J., Chang, M.-W., Lee, K., Google, K. T., & Language, A. I. (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. *Proceedings of the 2019 Conference of the North*, 4171–4186. <https://doi.org/10.18653/V1/N19-1423>
- Efron, B. (1979). Bootstrap Methods: Another Look at the Jackknife. <https://doi.org/10.1214/Aos/1176344552>, 7(1), 1–26. <https://doi.org/10.1214/AOS/1176344552>
- Gjurkovic, M., Karan, M., Vukojevic, I., Bosnjak, M., & Snajder, J. (2021). PANDORA Talks: Personality and Demographics on Reddit. *SocialNLP 2021 - 9th International Workshop on Natural Language Processing for Social Media, Proceedings of the Workshop*, 138–152. <https://doi.org/10.18653/V1/2021.SOCIALNLP-1.12>
- Goldberg, L. R. (1990). An alternative “description of personality”: the big-five factor structure. *Journal of Personality and Social Psychology*, 59(6), 1216–1229. <https://doi.org/10.1037//0022-3514.59.6.1216>
- Gururangan, S., Marasovic, A., Swayamdipta, S., Lo, K., Beltagy, I., Downey, D., & Smith, N. A. (2020). Don’t Stop Pretraining: Adapt Language Models to Domains and Tasks. *Proceedings of the Annual Meeting of the Association for Computational Linguistics*, 8342–8360. <https://doi.org/10.18653/V1/2020.ACL-MAIN.740>
- Ji, S., Zhang, T., Ansari, L., Fu, J., Tiwari, P., & Cambria, E. (2022). *MentalBERT: Publicly Available Pretrained Language Models for Mental Healthcare* (pp. 7184–7190). <https://aclanthology.org/2022.lrec-1.778/>
- Jolly, M. (2017). *MBTI Myers-Briggs Personality Type Dataset*. Kaggle. <https://www.kaggle.com/datasets/datasnaek/mbti-type>
- Kong, S., & Sokolova, M. (2024). *Explainable Multi-Label Classification of MBTI Types*. <https://arxiv.org/pdf/2405.02349>

- Kosinski, M., Stillwell, D., & Graepel, T. (2013). Private traits and attributes are predictable from digital records of human behavior. *Proceedings of the National Academy of Sciences of the United States of America*, 110(15), 5802–5805. <https://doi.org/10.1073/PNAS.1218772110>
- Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., Levy, O., Lewis, M., Zettlemoyer, L., Stoyanov, V., & Allen, P. G. (2019). *RoBERTa: A Robustly Optimized BERT Pretraining Approach*. <https://arxiv.org/pdf/1907.11692>
- Mairesse, F., Walker, M. A., Mehl, M. R., & Moore, R. K. (2007). Using Linguistic Cues for the Automatic Recognition of Personality in Conversation and Text. *Journal of Artificial Intelligence Research*, 30, 457–500. <https://doi.org/10.1613/JAIR.2349>
- McCrae, R. R., & Costa, P. T. (1989). Reinterpreting the Myers-Briggs Type Indicator from the perspective of the five-factor model of personality. *Journal of Personality*, 57(1), 17–40. <https://doi.org/10.1111/J.1467-6494.1989.TB00759.X>
- Myers, I. Briggs., H., M. M., Quenk, N. L., & Hammer, A. L. (1998). *MBTI manual : a guide to the development and use of the Myers-Briggs Type Indicator*. 420. [https://books.google.com/books/about/MBTI\\_Manual.html?hl=en&id=XPbRQwAACAAJ](https://books.google.com/books/about/MBTI_Manual.html?hl=en&id=XPbRQwAACAAJ)
- Park, G., Andrew Schwartz, H., Eichstaedt, J. C., Kern, M. L., Kosinski, M., Stillwell, D. J., Ungar, L. H., & Seligman, M. E. P. (2015). Automatic personality assessment through social media language. *Journal of Personality and Social Psychology*, 108(6), 934–952. <https://doi.org/10.1037/PSPP0000020>
- Pennebaker, J. W., Boyd, R. L., Jordan, K., & Blackburn, K. (2015). *The development and psychometric properties of LIWC2015*. <https://doi.org/10.15781/T29G6Z>
- Read, J., Pfahringer, B., Holmes, G., Frank, E., Brodley Read, C. J., Pfahringer, B., Holmes, G., Frank, E., & Read, J. (2011). Classifier chains for multi-label classification. *Machine Learning 2011* 85:3, 85(3), 333–359. <https://doi.org/10.1007/S10994-011-5256-5>
- Salton, G., & Buckley, C. (1988). Term-weighting approaches in automatic text retrieval. *Information Processing & Management*, 24(5), 513–523. [https://doi.org/10.1016/0306-4573\(88\)90021-0](https://doi.org/10.1016/0306-4573(88)90021-0)
- Schwartz, H. A., Eichstaedt, J. C., Kern, M. L., Dziurzynski, L., Ramones, S. M., Agrawal, M., Shah, A., Kosinski, M., Stillwell, D., Seligman, M. E. P., & Ungar, L. H. (2013). Personality, Gender, and Age in the Language of Social Media: The Open-Vocabulary Approach. *PLOS ONE*, 8(9), e73791. <https://doi.org/10.1371/JOURNAL.PONE.0073791>
- Tsoumakas, G., Katakis, I., & Vlahavas, I. (2011). Random k-labelsets for multilabel classification. *IEEE Transactions on Knowledge and Data Engineering*, 23(7), 1079–1089. <https://doi.org/10.1109/TKDE.2010.164>
- Verhoeven, B., Daelemans, W., & Plank, B. (2016). *Twisty: A Multilingual Twitter Stylogram Corpus for Gender and Personality Profiling* (pp. 1632–1637). <https://aclanthology.org/L16-1258/>
- Warner, B., Chaffin, A., Clavié, B., Weller, O., Hallström, O., Taghadouini, S., Gallagher, A., Biswas, R., Ladhak, F., Aarsen, T., Cooper, N., Adams, G., Howard, J., & Poli, I. (2025). Smarter, Better, Faster, Longer: A Modern Bidirectional Encoder for Fast, Memory Efficient, and Long Context Finetuning and Inference. *Proceedings of the Annual Meeting of the Association for Computational Linguistics*, 1, 2526–2547. <https://doi.org/10.18653/V1/2025.ACL-LONG.127>
- Zhang, M. L., & Zhou, Z. H. (2014). A Review on Multi-Label Learning Algorithms. *IEEE Transactions on Knowledge and Data Engineering*, 26(08), 1819–1837. <https://doi.org/10.1109/TKDE.2013.39>